

IDENTIFICATION

PRODUCT CODE: MAINDEC-08-DJMSA-A-D
PRODUCT NAME: 1-4K MS8-A MOS MEMORY TEST
DATE CREATED: SEPTEMBER 1974
MAINTAINER: PDP-8 DIAGNOSTICS
AUTHOR: MARK SANDLER

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1. ABSTRACT

THE 1-4K MS8-A MOS MEMORY TEST IS A PROGRAM THAT WILL TEST MOS MEMORIES FROM 1K UP TO 4K. IT CONSISTS OF AN ADDRESS SELECTION TEST, A FLOATING ONE'S AND ZEROS TEST AND A WORST CASE DATA TEST. THIS PROGRAM PROVIDES CPU-XOR, ACT-8/A, ACT-8/E AND STAND ALONE FRAME CAPABILITIES.

2. REQUIREMENTS

2.1 HARDWARE

THE FOLLOWING HARDWARE IS REQUIRED FOR THE EXECUTION OF THIS PROGRAM.

PROCESSOR(S): PDP8A

MEMORY: AT LEAST 1K OF MOS RAM

OPTIONS: MS8-A MOS RAM MEMORY

2.2 STORAGE

PROGRAM REQUIRES 4 PAGES OF A 1K RANGE.

2.3 PREREQUISITE SOFTWARE

PRIOR TO RUNNING THIS PROGRAM, THE PDP8A CENTRAL PROCESSOR TEST (MAINDEC-08-DJKKA) MUST HAVE BEEN SUCCESSFULLY RUN.

3. RESTRICTIONS

A. ASSUMPTION IS MADE THAT RAM WILL NEVER GO BEFORE ROM MEMORY.

B. THE 1K RAM IN WHICH THE DIAGNOSTIC IS LOADED WILL BE CONSIDERED THE LOWEST 1K SEGMENT WHEN TEST BEGINS.

C. THE PROGRAM WILL HAVE TO BE RELOADED WHENEVER THE USER WISHES TO CHANGE TO ANOTHER TEST ENVIRONMENT (STAND ALONE, CPU-XOR, ACT-8/A, ACT-8/E).

STANDARD TEST PROCEDURE

THE MODIFIED RIM LOADER IN 4.3 IS TOGGLED IN. THEN THE DIAGNOSTIC IS READ IN TO THE DESIRED 1K RANGE FROM THE RIM FORMAT TAPE THE PROGRAM STARTS AT LOCATION 200 OF THE CHOSEN RANGE. THE TOP HALF OF THIS RANGE AND ALL 1K SEGMENTS ABOVE IT WILL BE TESTED. THE ADDRESS OF THE LAST ADDRESS OF THE HIGHEST RANGE TO BE TESTED WILL BE IN LOCATION 23 OF THE CHOSEN LOWEST RANGE INITIALLY. AFTER THE 3 TESTS ARE COMPLETED THE PROGRAM WILL FLIP TO THE TOP HALF OF IT'S RANGE AND THE BOTTOM HALF AND ALL HIGHER RANGES WILL BE TESTED. THE PROGRAM WILL FLIP BACK AND THE PROCESS WILL CONTINUE FROM THE BEGINNING. ONE FULL PASS FOR 4K OF MEMORY REQUIRES ABOUT 23 SECONDS.

- A. THE DIAGNOSTIC IS IN RIM FORMAT. TOGGLE IN THE RIM LOADER MODIFIED AS DESCRIBED IN 4.2.
- B. READ IN THE 1-4K MS8-A MOS MEMORY TEST FROM THE TAPE. DEPOSIT THE LAST ADDRESS OF THE HIGHEST ADDRESS RANGE TO BE TESTED INTO LOCATION 23 OF THE RANGE THE DIAGNOSTIC WAS LOADED INTO. THE DEFAULT IS 1777 WHICH SIGNIFIES A 1K MEMORY STARTING AT LOCATION 0.
- C. SET SWITCH REGISTER TO LOCATION 200 OF THE RANGE THE DIAGNOSTIC WAS LOADED INTO. PRESS LOAD ADDRESS, SET SWITCH REGISTER ACCORDING TO 6.2. IN ADDITION TO THE ERROR RELATED SWITCHES IN 6.2 SET SW3 TO A ONE IF THE CPU IS TO HALT AFTER ONE FULL PASS. IF SW3=0 THE DIAGNOSTIC WILL LOOP INDEFINITELY.
- D. PRESS CLEAR AND THEN CONTINUE
- E. PROGRAM WILL HALT ON ERROR OR ON PASS COMPLETION IF SW3 IS UP. OTHERWISE IT WILL RUN INDEFINITELY.

4.1

CHANGING DEVICE IOT CODES

NOT APPLICABLE

MODIFIED RIM LOADER

IN ORDER TO LOAD THE DIAGNOSTIC INTO A
1K RANGE OTHER THAN 0-1777 THE MODIFIED
RIM LOADER IS USED OTHERWISE THE STANDARD
RIM LOADER IS SATISFACTORY,

LOW MOD. RIM LOADER
-----HIGH MOD. RIM LOADER

X753/	6032 KCC	6014	RCF
X754/	0375 AND X775	0375	AND X775
X755/	1377 TAD X777	1377	TAD X777
X756/	3376 DCA X776	3376	DCA X776
X757/	6031 KSF	6011	RSF
X760/	5357 JMP .-1	5357	JMP .-1
X761/	6036 KRB	6016	RCC
X762/	7106 CLL RTL	7106	CLL RTL
X763/	7006 RTL	7006	RTL
X764/	6031 SPA	7510	SPA
X765/	5367 JMP X757	5354	JMP X754
X766/	6034 RTL	7006	RTL
X767/	7420 KSF	6011	RSF
X770/	5367 JMP .-1	5367	JMP .-1
X771/	6034 KRS	6016	RCC
X772/	7420 SNL	7420	SNL
X773/	3776 DCA I X776	3776	DCA I X776
X774/	5353 JMP X753	5354	JMP X754
X775/	1777 /MASK	1777	/MASK
X776/	0	0	
X777/	0 /ADDRESS RANGE	0	/ADDRESS RANGE

THE USER WILL MODIFY X777 TO ADDRESS RANGE
DESIRED; (0000,2000,4000,6000),

5.

ERRORS

ALL ERRORS ARE REPORTED BY ERROR HALTS.

5.1

ERROR HALTS/INDICATIONS

ALL TESTS USE THE SAME ERROR REPORTER ROUTINE.
ERROR HALTS WILL OCCUR AT EITHER LOCATION 737 OR
LOCATION 1737, DEPENDING ON WHICH PORTION OF MOS RAM
THE PROGRAM HAS ENTERED.

ERRORS SHOULD BE HANDLED IN THE FOLLOWING SEQUENCE:

- A. AC WILL DISPLAY PC OF TEST IN ERROR.
- B. HIT CONTINUE
- C. AC WILL DISPLAY LOCATION BEING TESTED
- D. HIT CONTINUE
- E. AC WILL DISPLAY CONTENTS OF LOCATION
- F. HIT CONTINUE
- G. AC WILL DISPLAY EXPECTED CONTENTS.
- H. SET SWITCH REGISTER FOR RECOVERY AS DESIRED ACCORDING TO 6.2.

PROGRAM SWAP OR PROGRAM RELOCATION ERRORS WILL HALT AT EITHER LOCATION 264 OR LOCATION 1264. THIS TYPE OF ERROR IS FATAL AND INDICATES THAT THE MOS RAM IS FAULTY AND THAT THE PROGRAM SHOULD NOT BE CONTINUED.

5.2 ERROR PRINTOUTS

THERE ARE NO ERROR PRINTOUTS

6. SWITCH REGISTER SETTINGS

IF YOU DO NOT HAVE A FRONT PANEL DEPOSIT THE DESIRED SWITCH REGISTER CONTENTS IN LOCATION 21 OF THE RANGE THE PROGRAM IS USING BY OS/8 ODT ETC. IF YOU DO HAVE A FRONT PANEL DEPOSIT 4000 IN LOCATION 21 OF THAT SAME 1K SEGMENT.

6.1 NORMAL OPERATING SWITCHES

SW3=0 RUN DIAGNOSTIC INDEFINITELY
SW3=1 HALT ON PASS COMPLETION

6.2 ERROR RELATED SWITCHES

SW0=0 HALT ON ERROR
SW0=1 INHIBIT ERROR HALT

SW1=0 DO NEXT LOCATION AFTER ERROR
SW1=1 LOOP ON ERROR

SW2=0 DO ALL 3 TESTS
SW2=1 LOOP ON TEST

7. REVISIONS

NONE

PROGRAM DESCRIPTION

THIS DIAGNOSTIC FITS IN 4 PAGES OF A 1K SEGMENT. IT WILL TEST THE UPPER 4 PAGES PLUS UP TO 3K OF RAM MEMORY ABOVE THE 1K SEGMENT. THEN THE PROGRAM RELOCATES TO THE UPPER 4 PAGES AND TESTS THE LOWER 4 PAGES PLUS UP TO 3K OF RAM MEMORY ABOVE. THE PROGRAM THEN RELOCATES BACK TO REPEAT THE CYCLE CONTINUOUSLY. THE MODIFIED RIM LOADER IN 4.2 LOADS THE DIAGNOSTIC INTO THE DESIRED 1K SEGMENT. THIS BECOMES THE LOWEST 1K SEGMENT TO BE TESTED. BEFORE RUNNING THE PROGRAM THE LAST ADDRESS TO BE TESTED IS DEPOSITED IN LOCATION 23. THE PROGRAM WRITES OVER THE RIM LOADER IN OPERATION, SO IT MUST BE TOGGLED IN AGAIN IF DESIRED AFTER THE PROGRAM RUNS 3 TESTS ON THE RAM MEMORY.

A. TEST1 ADDRESS SELECTION TEST

EACH LOCATION BEING TESTED HAS IT'S ADDRESS WRITTEN INTO IT'S SELF. THEY ARE THEN ALL TESTED SCANNING BACKWARDS. THEN SCANNING BACKWARDS EACH LOCATION HAS THE COMPLEMENT OF IT'S ADDRESS WRITTEN INTO ITSELF. THE MEMORY IS THEN TESTED SCANNING FORWARDS.

B. TEST2 FLOATING ONES AND ZEROS TEST

THROUGH EACH LOCATION BEING TESTED A WORD IS WRITTEN THEN TESTED. THE WORD CONSISTS OF THE TWELVE WAYS TO A 1 CAN BE FLOATED (FROM 1,2,4---4096) AND THE TWELVE WAYS A ZERO CAN BE FLOATED OR THE COMPLEMENT OF THE LAST 12 WORDS. THIS TEST CHECKS FOR INTRA-ADDRESS BIT SHORTS.

C. TEST3 WORST CASE DATA PATTERN TEST

1. WRITE WORST CASE DATA PATTERN THROUGH RAM AND TEST EACH LOCATION
2. REPEAT WITH COMPLEMENT OF PATTERN
3. VARY PATTERN AND CONTINUE AT STEP 1. AFTER A MAXIMUM OF 12 PATTERNS FALL OUT OF TEST, THE PATTERN WILL BE VARIED FROM X,X',X,X'... TO X,X,X',X'... TO X,X,X,X,X',X',X',X' TO ETC. THE LAST PATTERN IN THIS PROGRESSION WILL HAVE THE FIRST HALF OF THE RAM CONTAIN X AND THE REST CONTAIN X'. THE DEFAULT FOR THIS TEST IS X=2525 AND X'=5252.

D. ACT-8A CAPABILITY

ACT-8A HOOKS HAVE BEEN PROVIDED IN THE PROGRAM WHICH WILL REPLACE ALL ERROR HALTS WITH EXIT RETURNS TO HOST WHEN RUNNING PROGRAM ON THE ACT-8A SYSTEM.

E. CPU-XOR CAPABILITY

XOR HOOKS HAVE BEEN PROVIDED IN THE PROGRAM TO LOOP ON XOR ERRORS WHEN RUNNING ON THE CPU-XOR TESTER.

F. ACT-8E CAPABILITY

ACT-8E HOOKS HAVE BEEN PROVIDED IN THE PROGRAM TO ALLOW IT TO BE RUN ON THE ACT-8E SYSTEM.

9. FLOWCHARTS

NONE

10. PROGRAM LISTING

10-11-1944
The following information was received from the
Office of the Director of the Bureau of
Prisons, New York City, on October 11, 1944:

On October 11, 1944, the following information was received from the
Office of the Director of the Bureau of Prisons, New York City:

The following information was received from the
Office of the Director of the Bureau of Prisons, New York City, on October 11, 1944:

Very truly yours,
Director

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/1-4K MSB-A MOS MEMORY TEST
 /MAINDEC-09-DJMSA-A-L
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*10
 MQA=7501
 MQL=7421
 SWP=7521
 XRON=6170
 SKXR=6171
 STIP=6173
 XRT0=6176
 FLOPAT, 0
 FLMCNT, -1
 LEMN2, 0
 K2777, 0777
 FLTPAT, 0
 FLPAT, 4000
 FLCNT, -2

/TWO'S COMP. OF TEST PATTERN
 /TEST PATTERN FOR TST2
 /DO TWICE, ONCE WITH COMP.

*20
 SWR, 0
 OP1, 0
 OP2, 0
 MEMLM, 1777

/ROUTINE TO TEST EACH ADDRESS WITH FLOATING ONES AND ZEROS

TST2, 0
 200+ TAD I AMEMB2
 200+ DCA FLADR /BEGINNING ADDRESS
 200+ TAD I AMEMB2
 200+ TAD K0777
 200+ CMA
 200+ DCA LEMN2 /END OF FIRST SEGMENT ADDRESS
 200+ CMA
 200+ DCA FLMCNT
 200+ CLA CLL CMA RAL /TAD (-2)
 200+ DCA FLCNT
 200+ CLL CML RAR /TAD (4000)
 200+ DCA FLPAT /SAVE INITIAL PATTERN
 200+ TAD FLCNT
 200+ CLL CMA RAR /SET UP LINK FOR TEST,
 200+ TAD FLPAT
 MQL
 MQA
 CMA
 SNL
 SWP
 IAC
 200+ DCA FLOPAT /SAVE TWO'S COMP. OF TEST PATTERN,
 MQA
 200+ DCA FLTPAT /SAVE TEST PATTERN

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FL2, 200+ TAD FLTPAT
 200+ DCA I FLADR /STORE PATTERN IN LOC. TO BE TESTED.
 200+ TAD I FLADR
 200+ TAD FLOPAT
 200+ SNA CLA /WAS PATTERN STORED CORRECTLY?
 200+ JMP FL3
 200+ TAD FLTPAT /SAVE EXPECTED PATTERN,
 MQL
 200+ TAD FLADR /SAVE CURRENT ADDRESS.
 200+ JMS I AERR2
 200+ JMP FL2 /ERROR RETRY RETURN,
 200+ JMS I AREPT2 /CHECK FOR XOR ERROR,
 200+ JMP TST2+1 /ERROR RETURN
 200+ TAD FLPAT
 200+ CLL RAR
 200+ DCA FLPAT /ROTATE PATTERN
 200+ SNL /ARE WE OFF THE END?
 200+ JMP FL1 /NO DO NEXT PATTERN
 200+ ISZ FLCNT /RETEST ADDRESS WITH COMP. OF PATTERN?
 200+ JMP FL0
 200+ ISZ FLADR /BUMP ADDRESS POINTER
 200+ TAD FLADR
 200+ AND (377)
 200+ SNA CLA
 200+ JMS I AREPT2 /IT IS TIME TO REPORT TO AUTO ACCEPTANCE
 200+ TAD FLADR
 200+ TAD LEMN2
 200+ SZA CLA /IS SEGMENT FINISHED?
 200+ JMP FL00 /NO DO NEXT ADDRESS
 200+ ISZ FLMCNT /SKIP IF FIRST SEGMENT,
 200+ JMP FL4
 200+ TAD I AFLIP2
 200+ SPA /IS THERE A GAP TO CROSS?
 200+ CLA /DO NOT SKIP A 1000 WORD GAP.
 200+ TAD FLADR
 200+ DCA FLADR
 200+ TAD I AMEML2
 200+ CMA
 200+ DCA LEMN2 /SET TO END OF SECOND SEGMENT
 200+ JMP FL35
 200+ JMS MYLAS /GET SWR
 200+ RTL
 200+ SPA CLA /LOOP ON TEST?
 200+ JMP TST2+1
 200+ JMP I TST2 /RETURN

/ROUTINE TO GET SWR INTO AC

MYLAS, 0
 200+ TAD OP1 /GET OPTIONS
 200+ SPA CLA /SKIP ON NO FRONT PANEL
 200+ JMP +3
 200+ TAD SWR /GET PSEUDO SWR

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111 0137 5732      200+  JMP I  MYLAS
112 0140 7404      OSR
113 0141 5732      200+  JMP I  MYLAS
114
115
116
117 /ROUTINE TO INITIALIZE FOR BEGINNING OF PASS FOR TST3
118 /
119 0142 0000      MINT3, 0
120 0143 7240      CLA CMA
121 0144 3763      200+  DCA I  ANSMCT /SET UP TO PASS TWICE THROUGH MSET3
122 0145 1755      200+  TAD I  AMEMB2
123 0146 1213      200+  TAD  K0777
124 0147 7040      CMA
125 0150 3762      200+  DCA I  ALEMN /SET TO END OF FIRST SEGMENT,
126 0151 1755      200+  TAD I  AMEMB2
127 0152 3761      200+  DCA I  ANSADR /SET TO BEGINNING OF SEGMENT,
128 0153 5742      200+  JMP I  MINT3
129 0154 0000      FLADR, 0
130 /RELOCATABLE CONSTANTS
131 0155 0351      AMEMB2, MEMBG /START ADDRESS OF MEMORY
132 0156 0023      AMEML2, MEMLM
133 0157 0355      AFLIP2, FLIPCN
134 0160 0731      AERR2, ERR
135 0161 0761      ANSADR, NSADR
136 0162 0756      ALEMN, LEMN
137 0163 0757      ANSMCT, NSMCNT
138 0164 0334      AREPT2, REPT
139 0165 0000      0 /END RELOCATABLE CONSTANTS
140
141 0200 5200      *200
142 0201 6170      START, JMP  XRON  .+6 /FIRST XRON
143 0202 6173      STIP
144 0203 5201      JMP  .+2 /LOOP UNTIL READY
145 0204 7344      CLA CMA CLL RAL /TAD -2
146 0205 3353      DCA  XCNT
147 0206 4207      JMS  RELOC
148 0207 7402      RELOC, HLT /WILL CONTAIN IT'S ADRESS
149 0210 1355      TAD  FLIPCN
150 0211 0207      AND  RELOC
151 0212 3207      DCA  RELOC /SAVE ORIGIN
152 0213 1207      REL0, TAD  RELOC
153 0214 1377      TAD  (LNK0)
154 0215 3341      DCA  LNKPOS /POINT TO PAGE ZERO TABLE
155 0216 1741      REL1, TAD I  LNKPOS
156 0217 1207      TAD  RELOC
157 0220 3350      DCA  LNKIDX /POINT TO FIRST ENTRY IN TABLE
158 0221 1750      REL2, TAD I  LNKIDX /GET TAG
159 0222 7450      SNA
160 0223 5231      JMP  REL3 /IF ZERO DO NEXT PAGE
161 0224 0354      AND  RELAN /MAKE RELATIVE TO LOC 0
162 0225 1207      TAD  RELOC
163 0226 3750      DCA I  LNKIDX /FIX TAG
164 0227 2350      ISZ  LNKIDX /BUMP TAG POINTER
165

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166 0230 5221      JMP  REL2 /DO NEXT TAG ON PAGE
167 0231 2341      REL3, ISZ  LNKPOS //BUMP PAGE TABLE POINTER
168 0232 1341      TAD  LNKPOS
169 0233 7041      CIA
170 0234 1357      TAD  ALEND
171 0235 7640      SZA CLA /ARE ALL 4 PAGES DONE?
172 0236 5216      JMP  REL1 /DO NEXT PAGE,
173 0237 5771      GINT, JMP I  AINT /DO INITIALIZATION CODE,
174 0240 1355      TAD  FLIPCN
175 0241 7041      CIA
176 0242 1207      TAD  RELOC
177 0243 3351      DCA  MEMBG /SAVE MEMORY TEST BEGIN
178 0244 4762      RUN, JMS I  ATST1
179 0245 4763      JMS I  ATST2
180 0246 4764      JMS I  ATST3
181 0247 1355      TAD  FLIPCN
182 0250 7041      CIA
183 0251 3355      DCA  FLIPCN /SET UP TO FLIP PROGRAM LATER,
184 0252 1207      TAD  RELOC
185 0253 3352      DCA  RTEMP
186 0254 1351      TAD  MEMBG
187 0255 3207      DCA  RELOC /SET UP NEW ORIGIN
188 0256 1752      RL1, TAD I  RTEMP
189 0257 3751      DCA I  MEMBG
190 0260 1752      TAD I  RTEMP
191 0261 7041      CIA
192 0262 1751      TAD I  MEMBG
193 0263 7640      SZA CLA /IS INSTRUCTION CORRECT?
194 0264 7402      RLERR, HLT /ERROR IN RELOCATING,
195 0265 2352      ISZ  RTEMP
196 0266 2351      ISZ  MEMBG
197 0267 1352      TAD  RTEMP
198 0270 0376      AND  (0777)
199 0271 7640      SZA CLA /IS ALL CODE MOVED?
200 0272 5256      JMP  RL1
201 0273 1355      TAD  FLIPCN
202 0274 7700      SMA CLA /SKIP IF FULL PASS COMPLETED
203 0275 5306      JMP  SWCH
204 0276 4760      JMS I  AMYLAS /GET SWR
205 0277 0375      AND  (400) /HALT AT END OF PASS?
206 0300 7640      SZA CLA
207 0301 7402      HLT
208 0302 6170      X1, XRON /SECOND OR THIRD XRON FOR XOR,
209 0303 2756      ISZ I  AXCNT1
210 0304 5306      JMP  SWCH
211 0305 5767      CT1, JMP I  AOFF /DISABLE MODULE UNDER TEST
212 0306 1374      SWCH, TAD  (RELOC)
213 0307 1207      TAD  RELOC
214 0310 3352      DCA  RTEMP
215 0311 5752      JMP I  RTEMP /FLIP PROGRAM
216
217
218 0312 0000      MSET1, 0
219 0313 3352      DCA  RTEMP /SAVE DIRECTION OF SCAN
220 0314 2766      ISZ I  ASEMCT /FALL OUT ON THIRD PASS

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201 0315 5712 JMP I MSET1
202 0316 1355 TAD FLIPCN
203 0317 7510 SPA
204 0320 7200 CLA
205 0321 2352 ISZ RTEMP
206 0322 7041 CIA
207 0323 1765 TAD I ASELDR
208 0324 3765 DCA I ASELDR
209 0325 7501 MQA
210 0326 7041 CIA
211 0327 3761 DCA I ALEMN1
212 0330 1312 TAD MSET1
213 0331 1373 TAD (-14)
214 0332 3352 DCA RTEMP
215 0333 5752 JMP I RTEMP
216
217
218 0334 0000 REPT, 0
219 0335 6171 SKXR /SKIP ON XOR ERROR,
220 0336 2334 ISZ REPT
221 0337 5734 JMP I REPT
222 0340 5734 JMP I REPT /FOR AUTO ACCEPTANCE,
223
224 0341 0000 LNKPOS, 0 /POINTS TO A PAGE TABLE POINTER
225 0342 0155 LNK0, AMEM92 /PAGE 0 TABLE START
226 0343 0356 LNK1, AXCNT1 /PAGE 1 TABLE START
227 0344 0565 LNK2, AFLIP1 /PAGE 2 TABLE START
228 0345 0766 LNK3, AMEM93 /PAGE 3 TABLE START
229 0346 1126 LNK4, AMA1 /PAGE 4 TABLE START
230 0347 1257 LNK5, AA-LEND /PAGE 5 TABLE START
231 0350 0000 LNKIDX, 0 /INDIVIDUAL TABLE POINTER
232 0351 0000 MEMBG, 0 /BEG. OF MEMORY FOR TEST
233 0352 0000 RTEMP, 0
234 0353 0000 XCNT, 0 /XOR COUNTER,
235 0354 1777 RELAN, 1777 /RELOCATION MASK
236 0355 7000 FLIPCN, -1000 /CONSTANT TO FLIP PROGRAM,
237 /THESE CONSTANTS WILL BE RELOCATED
238
239 0356 0353 AXCNT1, XCNT
240 0357 0350 ALEND, LNK5+1
241 0360 0132 AMYLAS, MYLAS /ADDRESS OF ROUTINE TO GET SWR
242 0361 0564 ALEMN1, LEMN1
243 0362 0400 ATST1, TST1 /ADDRESS SELECTION TEST,
244 0363 0024 ATST2, TST2 /FLOATING ONES AND ZEROS TEST,
245 0364 0600 ATST3, TST3 /DATA TEST,
246 0365 0554 ASELDR, SELADR
247 0366 0562 ASEMCR, SEMCNT
248 0367 0544 AOFF, OFF
249 0370 0731 AERR, ERR
250 0371 1000 AINT, INT
251 0372 0000 0
252 0373 7764
253 0374 0213
254 0375 0400
255 0376 0777

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276 0377 0342 *400
277 0400 /ROUTINE TO DO ADDRESS SELECTION TEST
278
279 0400 0000 TST1, 0
280 0401 7344 CLA CMA CLL RAL /TAD(-2)
281 0402 3357 DCA SELCNT
282 0403 1377 TAD (NOP)
283 0404 3226 DCA SWL1
284 0405 1341 TAD LCIA
285 0406 3251 DCA SWL2
286 0407 1772 TAD I AMEMB1
287 0410 3355 DCA SELOP1 /SAVE BEGIN ADDRESS
288 0411 1766 TAD I AMEML1
289 0412 3356 DCA SELOP2 /SAVE END ADDRESS
290 0413 7001 LIAC, IAC
291 0414 3360 DCA SELINC /INIT, INC,
292 0415 7040 LCMA, CMA
293 0416 3361 DCA SELDEC /INIT, DECR,
294 0417 1226 SEL0, TAD SWL1
295 0420 3256 DCA SWL3
296 0421 1355 TAD SELOP1
297 0422 3354 DCA SELADR /SAVE BEG, ADDRESS
298 0423 1360 TAD SELINC
299 0424 4325 JMS MINT1
300 0425 1354 SEL1, TAD SELADR
301 0426 7402 SWL1, HLT
302 0427 3754 DCA I SELADR /STORE ADDRESS IN SELF
303 0430 4774 JMS I AREPT1 /CHECK FOR XOR ERROR,
304 0431 5201 JMP TST1+1 /ERROR RETURN,
305 0432 1354 TAD SELADR
306 0433 1364 TAD LEMN1
307 0434 7650 SNA CLA /IS FIRST SEGMENT FINISHED?
308 0435 5242 JMP SEL15
309 0436 1354 TAD SELADR
310 0437 1360 TAD SELINC
311 0440 3354 DCA SELADR /BUMP ADDRESS
312 0441 5225 JMP SEL1
313 0442 1356 SEL15, TAD SELOP2
314 0443 7421 MQL
315 0444 1361 TAD SELDEC
316 0445 4771 JMS I AMSET1 /BUMP LEMN
317 0446 1361 TAD SELDEC
318 0447 4325 JMS MINT1
319 0450 1754 SEL3, TAD I SELADR
320 0451 7402 SWL2, HLT
321 0452 1354 TAD SELADR
322 0453 7650 SNA CLA /IS THERE AN ERROR
323 0454 5263 JMP SEL35
324 0455 1354 TAD SELADR /SAVE EXPECTED PATTERN
325 0456 7402 SWL3, HLT
326 0457 7421 MQL
327 0460 1354 TAD SELADR /SAVE CURRENT ADDRESS
328 0461 4767 JMS I AERR1
329 0462 5250 JMP SEL3 /ERROR RETRY ADDRESS

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330 0463 4774 SEL35, JMS I AREPT1 /CHECK FOR XOR ERROR,
331 0464 5201 JMP TST1+1 /ERROR RETURN.
332 0465 1354 TAD SELADR
333 0466 1364 TAD LEMN1
334 0467 7650 SNA CLA /IS FIRST SEGMENT FINISHED?
335 0470 5275 JMP SEL4
336 0471 1354 TAD SELADR
337 0472 1361 TAD SELDEC
338 0473 3354 DCA SELADR /BUMP ADDRESS
339 0474 5250 JMP SEL3
340 0475 1355 SEL4, TAD SELOP1
341 0476 7421 MQL
342 0477 1360 TAD SELINC
343 0500 4771 JMS I AMSET1 /BUMP LEMN1
344 0501 7201 CLA IAC /TAD(1)
345 0502 3361 DCA SELDEC /SCAN IN OPP. DIRECTION
346 0503 7240 CLA CMA /TAD(-1)
347 0504 3360 DCA SELINC /SCAN IN OPP. DIRECTION
348 0505 1215 TAD LCMA
349 0506 3226 DCA SWL1
350 0507 1213 TAD LIAC
351 0510 3251 DCA SWL2
352 0511 1766 TAD I AMEML1
353 0512 3355 DCA SELOP1
354 0513 1772 TAD I AMEMB1
355 0514 3356 DCA SELOP2 /SWAP END POINTS.
356 0515 2357 ISZ SELCNT
357 0516 5217 JMP SEL0 /DO TEST IN OPP. DIRECTION
358 0517 4774 MA1, JMS I AREPT1 /TIME TO REPORT IN A GOOD PASS.
359
360 0520 4770 JMS I AMLAS1 /GET SWR
361 0521 7006 RT
362 0522 7710 SPA CLA /LOOP ON TEST?
363 0523 5201 JMP TST1+1
364 0524 5600 JMP I TST1 /RETURN
365 /
366
367 /ROUTINE TO SET LOCAL END OF SEGMENT POINTER
368 /
369 /ROUTINE TO INITIALIZE FOR TST1
370 /
371
372 0525 0000 MINT1, 0
373 0526 3363 DCA TMP2
374 0527 7240 MK1, CLA CMA
375 0530 3362 DCA SEMCNT
376 0531 2363 ISZ TMP2
377 0532 5337 JMP MNT2
378 0533 1765 MK2, TAD I AFLIP1
379 0534 7510 SPA
380 0535 7200 CLA
381 0536 7001 IAC
382 0537 1353 MNT2, TAD K0777A
383 0540 1772 TAD I AMEMB1
384 0541 7041 LCIA, CIA

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385 0542 3364 DCA LEMN1
386 0543 5725 JMP I MINT1
387 /
388
389 /ISSUE TIMEOUT FOR XOR AND EXIT.
390
391 0544 1776 OFF, TAD I (START+1)
392 0545 1773 TAD I ARELOC /RELOCATE START POINTER.
393 0546 3363 DCA TMP2
394 0547 6176 XRT0 /TIMEOUT
395 0550 6173 STIP
396 0551 5763 JMP I TMP2 /DO NEXT MODULE.
397 0552 5354 JMP .+2 /WAIT TILL MODULE IS OFF.
398
399 /CONSTANTS
400
401 0553 0777 K0777A, 0777
402 0554 0000 SELADR, 0 /CURRENT ADDRESS IN TST1.
403 0555 0000 SELOP1, 0 /DELIMITS SCAN RANGE
404 0556 0000 SELOP2, 0 /DELIMITS SCAN RANGE
405 0557 7776 SELCNT, -2 /DO TEST TWICE ,ONCE IN EACH DIRECTION.
406 0560 0001 SELINC, 1
407 0561 7777 SELDEC, -1
408 0562 0000 SEMCNT, 0
409 0563 0000 TMP2, 0 /TEMPORARY
410 0564 0000 LEMN1, 0 /LOCAL END OF MEMORY.
411
412 /RELOCATABLE CONSTANTS
413
414 0565 0355 AFLIP1, FLIPCN
415 0566 0023 AMEML1, MEMLM
416 0567 0731 AERR1, ERR
417 0570 0132 AMLAS1, MYLAS
418 0571 0312 AMSET1, MSET1
419 0572 0351 AMEMB1, MEMBG
420 0573 0207 ARELOC, RELOC
421 0574 0334 AREPT1, REPT
422 0575 0000 0 /END OF CONSTANTS
423
424 0576 0201
425 0577 7000
426 0600
427 /THIS ROUTINE CHECKS FOR INTER-ADDRESS DATA BIT SHORTS
428 / AND ATTEMPTS TO SET UP A WORST CASE DATA PATTERN
429
430 0600 0000 TST3, 0
431 0601 7301 CLA CLL IAC /TAD (1)
432 0602 3362 DCA NSMASK /INITIAL MASK
433 0603 7344 NS0, CLA CLL CMA RAL /TAD (-2)
434 0604 3360 DCA NSCNT
435 0605 1363 NS1, TAD NSPAT1
436 0606 7104 CLL RAL
437 0607 7001 IAC
438 0610 3364 DCA NSPAT2 /SET UP COMPLEMENT
439 0611 1363 TAD NSPAT1

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439 0612 7040 CMA
440 0613 3363 DCA NSPAT1
441 0614 4772 JMS I AMINT3
442 0615 1361 NS2, TAD NSADR
443 0616 0362 AND NSMASK
444 0617 7640 SZA CLA /CHOOSE NSPAT1 OR NSPAT2 + NSPAT1
445 0620 1364 TAD NSPAT2
446 0621 1363 TAD NSPAT1
447 0622 3761 DCA I NSADR
448 0623 4773 NSX1, JMS I AREPT3 /STORE CHOSEN WORD
449 0624 5601 JMP I TST3+1 /CHECK FOR XOR ERROR,
450 0625 2361 ISZ NSADR /ERROR RETURN,
451 0626 1361 NS3, TAD NSADR
452 0627 1356 TAD LEMN
453 0630 7640 SZA CLA /ARE WE AT THE END?
454 0631 5215 JMP NS2
455 0632 4313 JMS MSET3
456 0633 5226 JMP NS3 /DO NEXT ADR
457 0634 4772 JMS I AMINT3
458 0635 1361 NS4, TAD NSADR
459 0636 0362 AND NSMASK
460 0637 7640 SZA CLA /CHOOSE COMP. OF PATTERN
461 0640 1364 TAD NSPAT2
462 0641 1363 TAD NSPAT1
463 0642 7041 CIA
464 0643 1761 TAD I NSADR
465 0644 7650 SNA CLA /IS IT CORRECT?
466 0645 5257 JMP NS5
467 0646 1361 TAD NSADR
468 0647 0362 AND NSMASK
469 0650 7640 SZA CLA /CHOOSE PAT.
470 0651 1364 TAD NSPAT2
471 0652 1363 TAD NSPAT1
472 0653 7421 MQL
473 0654 1361 TAD NSADR /SAVE EXPECTED PAT,
474 0655 4331 JMS ERR /SAVE CURRENT ADDRESS,
475 0656 5235 JMP NS4 /ERROR RETRY RETURN,
476 0657 4773 NS5, JMS I AREPT3 /CHECK FOR XOR ERROR,
477 0660 5201 JMP TST3+1
478 0661 2361 ISZ NSADR
479 0662 1361 NS6, TAD NSADR
480 0663 1356 TAD LEMN
481 0664 7640 SZA CLA /ARE WE AT THE END?
482 0665 5235 JMP NS4
483 0666 4313 JMS MSET3
484 0667 5262 JMP NS6
485 0670 2360 ISZ NSCNT /DO TWICE
486 0671 5205 JMP NS1
487 0672 1766 TAD I AMEMB3
488 0673 7041 CIA
489 0674 1767 TAD I AMEML3
490 0675 7170 CLL CML CMA RAR /DIVIDE BY =2
491 0676 1362 TAD NSMASK
492 0677 7700 SMA CLA
493 0700 5326 JMP NS7
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494 0701 1362 TAD NSMASK
495 0702 7104 RA_ CLL /ROTATE MASK
496 0703 3362 DCA NSMASK
497 0704 4773 MA3, JMS I AREPT3 /REPORT A GOOD PASS TO AUTO ACCEPTANCE,
498 0705 5203 JMP NS0
499 0706 4770 NS7, JMS I AMLAS3
500 0707 7006 RT_
501 0710 7710 SPA CLA /LOOP ON TEST?
502 0711 5201 JMP TST3+1
503 0712 5600 JMP I TST3
504
505 /ROUTINE TO BRIDGE GAP IN MEMORY FOR TST3
506
507 0713 0000 MSET3, 0
508 0714 2357 ISZ NSMCNT
509 0715 5327 JMP MS31
510 0716 1771 TAD I AFLIP3
511 0717 7510 SPA
512 0720 7200 CLA
513 0721 1361 TAD NSADR
514 0722 3361 DCA NSADR
515 0723 1767 TAD I AMEML3
516 0724 7040 CMA
517 0725 3356 DCA LEMN
518 0726 5713 JMP I MSET3
519 0727 2313 MS31, ISZ MSET3
520 0730 5713 JMP I MSET3
521
522 /ERROR REPORTER ROUTINE
523
524 0731 0000 ERR, 0
525 0732 3365 DCA ERRTMP
526 0733 4770 JMS I AMLAS3 /SAVE CURRENT ADDRESS
527 0734 7710 SPA CLA /GET SWR
528 0735 5351 JMP ERR1 /INHIBIT ERROR HALT?
529 0736 1331 TAD ERR
530 0737 7402 HLT
531 0740 7200 CLA /DISPLAY PC
532 0741 1365 TAD ERRTMP
533 0742 7402 HLT
534 0743 7200 CLA /DISPLAY CURRENT ADDRESS
535 0744 1765 TAD I ERRTMP /DISPLAY CONTENTS OF CURRENT ADDRESS
536 0745 7402 HLT
537 0746 7701 CLA MQA
538 0747 7402 HLT /DISPLAY EXPECTED CONTENTS
539 0750 7200 CLA
540 0751 4770 ERR1, JMS I AMLAS3
541 0752 7004 RA_
542 0753 7700 SMA CLA /LOOP ON ERROR?
543 0754 2331 ISZ ERR
544 0755 5731 JMP I ERR
545
546 /CONSTANTS
547
548 0756 0000 LEMN, 0 /LOCAL END OF MEMORY
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549 0757 7777 NSMCNT, -1
550 0763 7776 NSCNT, -2
551 0761 0000 NSADR, 0
552 0762 0000 NSMASK, 0
553 0763 2525 NSPAT1, 2525
554 0764 2525 NSPAT2, 2525
555 0765 0000 ERRTMP, 0
556
557 /RELOCATABLE CONSTANTS
558 0766 0351 AMEMB3, MEMBG
559 0767 0023 AMEML3, MEMLM
560 0770 0132 AMLAS3, MYLAS
561 0771 0355 AFLIP3, FLIPCN
562 0772 0142 AMINT3, MINT3
563 0773 0334 AREPT3, REPT
564 0774 0000
565 1000
566 *1000
567 /THIS CODE IS FOR INITIALIZATION OF LOCATION 23 FOR AUTO-ACCEPTANCE
568 /FROM MEMORY SIZE IN LOCATION 21 AND THE INSTRUCTION FIELD.
569
570 1000 1751 INT, TAD I AOP2
571 1001 7700 SMA CLA
572 1002 5254 JMP NAUTO
573 1003 1750 TAD I AOP1
574 1004 0377 AND (37)
575 1005 3352 DCA ITMP
576 1006 6224 RIF
577 1007 7110 CLL RAR
578 1010 7041 CIA
579 1011 1352 TAD ITMP
580 1012 3352 DCA ITMP
581 1013 1352 TAD ITMP
582 1014 0376 AND (34)
583 1015 7650 SNA CLA
584 1016 5221 JMP ,+3
585 1017 7325 CLA CLL CML IAC RAL
586 1020 7410 SKP
587 1021 1352 TAD ITMP
588 1022 7110 CLL RAR
589 1023 7012 RTI
590 1024 1375 TAD (1777)
591 1025 3746 DCA I AMEML4
592
593 /SET UP ROUTINE TO REPORT A GOOD PASS.
594
595
596 1026 1374 TAD (CIF 70)
597 1027 3740 DCA I AREPT4
598 1030 2340 ISZ AREPT4
599 1031 1373 TAD (REPT&177+4603) /JMS I REPT+3
600 1032 3740 DCA I AREPT4
601 1033 2340 ISZ AREPT4
602 1034 1372 TAD (6500)
603 1035 3740 DCA I AREPT4

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604 1036 1371 TAD (ERR&177+1200)
605 1037 3744 DCA I AERR4
606 1040 2344 ISZ AERR4
607 1041 1374 TAD (CIF 70)
608 1042 3744 DCA I AERR4
609 1043 2344 ISZ AERR4
610 1044 1370 TAD (ERR&177+5605) /JMS I ERR+5
611 1045 3744 DCA I AERR4
612 1046 2344 ISZ AERR4
613 1047 1367 TAD (6520)
614 1050 3744 DCA I AERR4
615 1051 1366 TAD (AERR&177+4600)
616 1052 3736 DCA I ARLERR
617
618 1053 5262 JMP CHKXOR
619 1054 1365 NAUTO, TAD (NOP)
620 1055 3726 DCA I AMA1
621 1056 1364 TAD (FL35&177+5200) /JMS FL35
622 1057 3727 DCA I AMA2
623 1060 1365 TAD (NOP)
624 1061 3730 DCA I AMA3
625 1062 1747 CHKXOR, TAD I AXCNT
626 1063 7640 SEA CLA
627 1064 5301 JMP LFIX
628 1065 1363 TAD (SKP)
629 1066 3731 DCA I AX1
630 1067 1363 TAD (SKP)
631 1070 3732 DCA I AFL3
632 1071 1363 TAD (SKP)
633 1072 3733 DCA I ASLX1
634 1073 1363 TAD (SKP)
635 1074 3741 DCA I ASEL35
636 1075 1363 TAD (SKP)
637 1076 3742 DCA I ANSX1
638 1077 1363 TAD (SKP)
639 1100 3743 DCA I ANS5
640
641 /CHECK IF ONLY 1K OF MEMORY IS PRESENT AND FIX MINT1 IF NECESSARY.
642
643 1101 7331 LFIX, CLA CLL CML IAC RAR
644 1102 0745 AND I AMEMB4
645 1103 1375 TAD (1777)
646 1104 7041 CIA
647 1105 1746 TAD I AMEML4
648 1106 7640 SEA CLA
649 1107 5325 JMP NXINT
650 1110 1365 TAD (NOP)
651 1111 3734 DCA I AMK1
652 1112 1362 TAD (AMEMB1&177+1600)
653 1113 3735 DCA I AMK2
654 1114 2335 ISZ AMK2
655 1115 1361 TAD (K0777A&177+1200)
656 1116 3735 DCA I AMK2
657 1117 2335 ISZ AMK2
658 1120 1360 TAD (SELADR&177+3200)

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659 1121 3735 DCA I AMK2
660 1122 2335 ISZ AMK2
661 1123 1363 TAD (SKP)
662 1124 3735 DCA I AMK2
663
664
665
666 1125 5737 NXINT, JMP I AINT2 /JMP TO NEXT INITIALIZATION PAGE
667
668
669 /RELOCATABLE CONSTANTS START HERE,
670
671 1126 0517 AMA1, MA1
672 1127 0105 AMA2, MA2
673 1130 0704 AMA3, MA3
674 1131 0302 AX1, X1
675 1132 0070 AFL3, FL3
676 1133 0430 ASLX1, SLX1
677 1134 0527 AMK1, MK1
678 1135 0533 AMK2, MK2
679 1136 0264 ARLERR, RLERR
680 1137 1200 AINT2, INT2
681 1140 0335 AREPT4, REPT+1
682 1141 0463 ASEL35, SE_35
683 1142 0623 ANSX1, VSX1
684 1143 0657 ANS5, VS5
685 1144 0733 AERR4, ERR+2
686 1145 0351 AMEMB4, MEMBG
687 1146 0023 AMEML4, MEMLM
688 1147 0353 AXCNT, XCNT
689 1150 0021 AOP1, OP1
690 1151 0022 AOP2, OP2
691 1152 0000 ITMP, 0
692 1160 3354
693 1161 1353
694 1162 1772
695 1163 7410
696 1164 5306
697 1165 7000
698 1166 4770
699 1167 6520
700 1170 5736
701 1171 1331
702 1172 6500
703 1173 4737
704 1174 6272
705 1175 1777
706 1176 0034
707 1177 0037
708 1200
709
710
711
712 1200 1660 INT2, TAD I AOP2B

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*1200
 /THIS PAGE IS THE INITIALIZATION THAT DID NOT FIT ON THE LAST PAGE.
 /IT SETS UP THE ACT 8/E HOOKS AND CUTS THE UMBILICAL CORD BETWEEN
 / THESE TWO PAGES AND THE MAIN CODE

```

713 1201 7004 RA,
714 1202 7700 SMA CLA /SKIP IF ACT 8/E
715 1203 5233 JMP NFIX
716 1204 1660 TAD I AOP2B
717 1205 7006 RTL
718 1206 7700 SMA CLA /SKIP ON QUICK VERIFY
719 1207 1377 TAD (-33) /10 MINUTES RUN TIME
720 1210 1376 TAD (-1) /ONE PASS FOR Q.V.
721 1211 3661 DCA I AXCNT5 /SET UP ISZ LOOP
722 1212 1375 TAD (NOP)
723 1213 3663 DCA I AX1B
724 1214 7040 CMA /TAD (7777)
725 1215 3664 DCA I AMEML5 /SET UP FOR 4K OF MEMORY
726 1216 1270 TAD ADONE
727 1217 3665 DCA I AAOFF /SET UP COMPLETION RETURN TO ACT 8/E
728 1220 1374 TAD (AERR&177+4600) /TAD (JMS I AERR)
729 1221 3671 DCA I ARLER5 /REPORT TO ACT 8/E RELOC, ERR,
730
731
732 /MOVE IN CODE FOR ACT8/E ERROR REPORTER AND COMPLETION RETRN
733
734 1222 1666 ACLOOP, TAD I ACT8E /GET INSTRUCTION
735 1223 3672 DCA I AERR5 /MOVE IT IN PLACE
736 1224 2272 ISZ AERR5 /BUMP POINTERS
737 1225 2266 ISZ ACT8E
738 1226 1266 TAD ACT8E
739 1227 7041 CIA
740 1230 1267 TAD ACTND
741 1231 7640 SZA CLA /ARE ALL INSTRUCTIONS MOVED?
742 1232 5222 JMP ACLOOP
743
744
745 /REMOVE THESE PAGES FROM PAGE RELOCATION TABLE BEFORE
746 / THIS PAGE GOES BYE BYE.
747
748 1233 7344 NFIX, CLA CLL CMA RAL /TAD (-2)
749 1234 1657 TAD I AALEND
750 1235 3657 DCA I AALEND
751 1236 1375 TAD (NOP)
752 1237 3673 DCA I AGINT /REMOVE JMP TO THIS PAGE.
753 1240 1373 TAD (0777)
754 1241 3674 DCA I ARELAN /FIX UP MASK FOR SWAPING
755 1242 2662 ISZ I ALNK1 /DO NOT RELOCATE AXCNT1 ANY MORE.
756 1243 5673 JMP I AGINT /RETURN
757
758
759 /THE FOLLOWING IS CODE TO BE MOVED IN FOR ACT 8/E ERROR
760 / REPORTER AND COMPLETION RETURN HOOKS.
761
762 1244 6731 CT8E, 6731 /SET DIAGNOSTIC ERROR
763 1245 7410 SKP
764 1246 6734 6734 /SET PROGRAM DONE
765 1247 7240 CLA CMA /TAD (7777)
766 1250 3003 3003 /DCA 3
767 1251 7240 CLA CMA

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768	1252	3002	3002	/DCA 2
769	1253	1344	ERR&177+1213	/TAD ERR+13
770	1254	3001	3001	/DCA 1
771	1255	5001	5001	/JMP 1
772	1256	1256	CTND,	
773				
774			/THE FOLLOWING ARE RELOCATABLE CONSTANTS	
775				
776	1257	0357	AALEND, ALEND	
777	1260	0022	AOP2B, OP2	
778	1261	0353	AXCNT5, XCNT	
779	1262	0343	ALNK1, LNK1	
780	1263	0302	AX1B, X1	
781	1264	0023	AMEML5, MEMLM	
782	1265	0367	AAOFF, AOFF	
783	1266	1244	ACT8E, CT8E	
784	1267	1256	ACTND, CTND	
785	1270	0735	ADONE, ERR+4	
786	1271	0264	ARLER5, RLERR	
787	1272	0733	AERR5, ERR+2	
788	1273	0237	AGINT, GINT	
789	1274	0354	ARELAN, RELAN	
790	1275	0132	AMLAS5, MYLAS	
791	1276	0000	0	
792			\$	
793	1373	0777		
794	1374	4770		
795	1375	7000		
796	1376	7777		
797	1377	7745		
798	0177	0377		

0000	00000000	11111110	11111111	11111111	11111111	11111111	11111111	11111111
0100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	00000001
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111000
1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11100000	11111111	11111111
1200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111110
1300	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00011111
1400								
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7200
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7400
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7600
7700

AAEND	1257	AREPT1	0574	LNK0	0342	RLI	0256
AAOFF	1265	AREPT2	0164	LNK1	0343	RLERR	0264
ACLOOP	1222	AREPT3	0773	LNK2	0344	RTEMP	0352
ACTBE	1266	AREPT4	1140	LNK3	0345	RUN	0244
ACTND	1267	ARLER5	1271	LNK4	0346	SEL0	0417
ADONE	1270	ARLERR	1136	LNK5	0347	SEL1	0425
AERR	0370	ASEL35	1141	LNKIDX	0350	SEL15	0442
AERR1	0567	ASELOR	0365	LNKPOS	0341	SEL3	0450
AERR2	0160	ASEM02	0366	MA1	0517	SEL35	0463
AERR4	1144	ASLX1	1133	MA2	0105	SEL4	0475
AERR5	1272	ATST1	0362	MA3	0704	SELAOR	0554
AFL3	1132	ATST2	0363	MEMBG	0351	SELCNT	0557
AFLIP1	0565	ATST3	0364	MEMLM	0023	SELDEC	0561
AFLIP2	0157	AX1	1131	MINT1	0525	SELINC	0560
AFLIP3	0771	AX1B	1263	MINT3	0142	SELOP1	0555
AGINT	1273	AXCNT	1147	MK1	0527	SELOP2	0556
AIN	0371	AXCNT1	0356	MK2	0533	SEMCNT	0562
AIN2	1137	AXCNT5	1261	MNT2	0537	SKXR	6171
ALEMN	0162	CHXOR	1062	HQA	7501	SLX1	0430
ALEMN1	0361	CT1	0305	ML	7421	START	0200
AEND	0357	CTBE	1244	MS31	0727	STIP	6173
ALNK1	1262	CTND	1256	MSET1	0312	SWL1	0426
AMA1	1126	ERR	0731	MSET3	0713	SWL2	0451
AMA2	1127	ERR1	0751	MYLAS	0132	SWL3	0456
AMA3	1130	ERRTMP	0765	NAUTO	1054	SWP	7521
AMEMB1	0572	FL0	0037	NFIX	1233	SWR	0020
AMEMB2	0155	FL00	0035	NS0	0603	SWTCH	0306
AMEMB3	0766	FL1	0041	NS1	0605	TMP2	0563
AMEMB4	1145	FL2	0055	NS2	0615	TS1	0400
AMEML1	0566	FL3	0070	NS3	0626	TS2	0024
AMEML2	0156	FL35	0106	NS4	0635	TS3	0600
AMEML3	0767	FL4	0125	NS5	0657	X1	0302
AMEML4	1146	FLADR	0154	NS6	0662	XCNT	0353
AMEML5	1264	FLCNT	0016	NS7	0706	XRON	6170
AMINT3	0772	FLCPAT	0010	NSADR	0761	XRT0	6176
AMK1	1134	FLIPCN	0355	NSCNT	0760		
AMK2	1135	FLMCNT	0011	NSMASK	0762		
AMLAS1	0570	FLPAT	0015	NSMCNT	0757		
AMLAS3	0770	FLTPAT	0014	NSPAT1	0763		
AMLAS5	1275	GINT	0237	NSPAT2	0764		
AMSET1	0571	INT	1000	NSX1	0623		
AMYLAS	0360	INT2	1200	NXINT	1125		
ANS5	1143	ITMP	1152	OFF	0544		
ANSADR	0161	K0777	0013	OP1	0021		
ANSMCT	0163	K0777A	0553	OP2	0022		
ANSX1	1142	LCIA	0541	REL0	0213		
AOFF	0367	LCMA	0415	REL1	0216		
AOP1	1150	LEMN	0756	REL2	0221		
AOP2	1151	LEMN1	0564	REL3	0231		
AOP2B	1260	LEMN2	0012	RELAN	0354		
ARELAN	1274	LFIX	1101	RELOC	0207		
ARELOC	0573	LIAC	0413	REPT	0334		

ERRORS DETECTED: 0
 LINKS GENERATED: 0
 RUN-TIME: 5 SECONDS
 3K CORE USED

AALEND	250	749	750	776#					
AAOFF	727	782#							
ACLOOP	734#	742							
ACT8E	734	737	738	783#					
ACTND	740	784#							
ADONE	726	785#							
AERR	269#	615	728						
AERR1	328	416#							
AERR2	66	134#							
AERR4	605	606	608	609	611	612	614	685#	
AERR5	735	736	787#						
AFL3	631	675#							
AFLIP1	247	378	414#						
AFLIP2	88	133#							
AFLIP3	510	561#							
AGINT	752	756	788#						
AIN1	173	270#							
AIN2	666	680#							
ALENN	125	136#							
ALENN1	231	262#							
ALEND	170	260#	776						
ALNK1	755	779#							
AMA1	249	620	671#						
AMA2	622	672#							
AMA3	624	673#							
AMEMB1	286	354	383	419#	652				
AMEMB2	32	34	122	126	131#	245			
AMEMB3	248	487	558#						
AMEMB4	644	686#							
AMEML1	288	352	415#						
AMEML2	93	132#							
AMEML3	489	515	559#						
AMEML4	591	647	687#						
AMEML5	725	781#							
AMINT3	441	457	562#						
AMK1	651	677#							
AMK2	653	654	656	657	659	660	662	678#	
AMLAS1	360	417#							
AMLAS3	499	526	540	560#					
AMLAS5	790#								
AMSET1	316	343	418#						
AMYLAS	204	261#							
ANS5	639	684#							
ANSADR	127	135#							
ANSMCT	121	137#							
ANSX1	637	683#							
AOFF	211	268#	782						
AOP1	573	689#							
AOP2	570	690#							
AOP2B	712	716	777#						
ARELAN	754	789#							
ARELOC	392	420#							
AREPT1	303	330	358	421#					

AREPT2	68	81	138#															
AREPT3	448	476	497	563#														
AREPT4	597	598	600	601	603	681#												
ARLER5	722	786#																
ARLERR	616	679#																
ASEL35	635	682#																
ASELDR	227	228	266#															
ASEMCZ	222	267#																
ASLX1	633	676#																
ATST1	178	263#																
ATST2	179	264#																
ATST3	180	265#																
AX1	629	674#																
AX1R	723	780#																
AXCNT	625	688#																
AXCNT1	209	246	259#															
AXCNT5	721	778#																
CHKXDR	618	625#																
CT1	211#																	
CT8E	762#	783																
CTND	772#	784																
ERR	134	269	416	474	524#	529	543	544	604	610	685	769	785	787				
ERR1	528	540#																
ERRTMP	525	532	535	555#														
FL0	42#	76																
FL00	40#	85																
FL1	44#	74																
FL2	56#	67																
FL3	62	68#	675															
FL35	82#	96	621															
FL4	87	97#																
FLADR	33	57	59	65	77	78	82	91	92	129#								
FLCNT	21#	41	44	75														
FLCPAT	15#	53	60															
FLIPCEN	133	150	174	181	183	201	222	256#	414	561								
FLMCNT	16#	39	86															
FLPAT	20#	43	46	70	72													
FLTPAT	19#	55	56	63														
GINT	173#	788																
INT	270	570#																
INT2	680	712#																
ITMP	575	579	580	581	587	691#												
K0777	18#	35	123															
K0777A	382	401#	655															
LCIA	284	384#																
LCMA	292#	348																
LEMN	136	452	480	517	548#													
LEMN1	262	306	333	385	410#													
LEMN2	17#	37	83	95														
LFIX	627	643#																
LIAC	290#	350																
LNK2	154	245#																
LNK1	246#	779																

LNK2	247#																	
LNK3	248#																	
LNK4	249#																	
LNK5	250#	260																
LNKIDX	158	159	164	165	251#													
LNKPOS	155	156	167	168	244#													
MA1	358#	671																
MA2	81#	672																
MA3	497#	673																
MEMBG	131	177	186	189	192	196	252#	419	558	686								
MEMLM	27#	132	415	559	687	781												
MINT1	299	318	372#	386														
MINT3	119#	128	562															
MK1	374#	677																
MK2	378#	678																
MNT2	377	382#																
MQA	8#	48	54	229	537													
SQL	9#	47	64	314	326	341	472											
MS31	509	519#																
MSET1	218#	221	232	418														
MSET3	455	483	507#	518	519	520												
MYLAS	97	106#	111	113	261	417	560	790										
NAUTO	572	619#																
NFIX	715	748#																
NS0	432#	498																
NS1	434#	486																
NS2	442#	454																
NS3	451#	456																
NS4	458#	475	482															
NS5	466	476#	684															
NS6	479#	484																
NS7	493	499#																
NSADR	135	442	447	450	451	458	464	467	473	478	479	513	514	551#				
NSCNT	433	485	550#															
NSMASK	431	443	459	468	491	494	496	552#										
NSMCNT	137	508	549#															
NSPAT1	434	438	440	446	462	471	553#											
NSPAT2	437	445	461	470	554#													
NSX1	448#	683																
NXINT	649	666#																
OFF	268	391#																
OP1	25#	107	689															
OP2	26#	690	777															
REL0	153#	212																
REL1	156#	172																
REL2	159#	166																
REL3	161	167#																
RELAN	162	255#	789															
RELOC	148	149#	151	152	153	157	163	176	184	187	213	420						
REPT	138	238#	240	241	242	421	563	599	681									
RL1	188#	200																
RLERR	194#	679	786															
RTMP	185	188	190	195	197	214	215	219	225	234	235	253#						

[illegible]